

Predicting DNA Methylation Status Using Machine Learning

Background

DNA methylation is an important biological process that regulates gene expression without changing the DNA sequence itself. Abnormal methylation patterns are often associated with various diseases, including cancer and genetic disorders. Understanding methylation patterns can help researchers gain deeper insights into gene regulation and biological mechanisms.

However, experimentally determining methylation status can be time-consuming and costly. Therefore, machine learning offers an alternative approach by predicting methylation status from existing genomic features.

Objective

The objective of this project is to build a classification model capable of predicting whether a genomic region is methylated or unmethylated based on several genomic characteristics.

Dataset

The dataset contains 1,000 genomic observations and includes the following features:

1. CpG Density
2. Genomic Location

3. Regulatory Score
4. Conservation Score
5. Target Variable:
6. Methylation Status (0 = Unmethylated, 1 = Methylated)

Methodology

The project follows a standard data science workflow:

1. Data Understanding and Exploration
 - Dataset inspection
 - Missing value checking
 - Statistical summary
 - Correlation analysis
2. Data Preparation
 - Feature-target separation
 - Train-test split
 - Feature scaling using StandardScaler
3. Model Development
 - Logistic Regression
 - Decision Tree Classifier
 - Random Forest Classifier
4. Model Evaluation
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix

Results

The Logistic Regression model achieved an accuracy of 98.5% on the test dataset. The model successfully classified most methylated and unmethylated samples with very few misclassifications.

This high performance indicates that the selected genomic features contain strong predictive information regarding DNA methylation status.

Key Findings

1. Genomic characteristics are highly informative for predicting methylation patterns.
2. Machine learning models can effectively distinguish between methylated and unmethylated regions.
3. Feature importance analysis reveals which genomic factors contribute most significantly to predictions.

Conclusion

This project demonstrates how machine learning can be applied in bioinformatics to predict DNA methylation status efficiently. The results suggest that genomic features can serve as strong predictors, providing a foundation for future studies involving larger biological datasets and more advanced predictive models.